Ocular Rosacea

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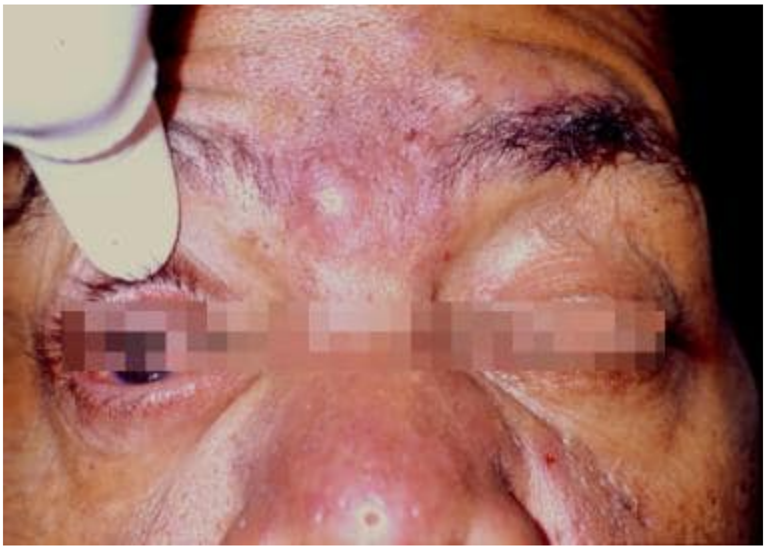
Overview

Background

Rosacea is a common inflammatory dermatologic condition that affects the midface and eyes.[1, 2, 3, 4, 5, 6, 7] Although the etiology of rosacea is not fully understood, an augmented response of the innate immune system and neurovascular pathways to certain triggers are considered to be major factors in the chronic inflammatory process associated with this condition.[8] The nose, cheeks, forehead, chin, and glabella are the most commonly affected areas. Clinical features include flushing, telangiectasias, erythema, papules and pustules, and rhinophyma.[1, 2, 3] More than 50% of patients with rosacea have ocular manifestations, and ocular findings may be the first manifestation of rosacea in some patients.

See the image below.



Typical dermatologic findings of rosacea, including midfacial papules, pustules, and rhinophyma.

Manifestations of ocular rosacea range from minor irritation, foreign body sensation, dryness, and blurry vision to severe ocular surface disruption and inflammatory keratitis. Patients frequently describe a gritty feeling, and they commonly experience Blepharitis and conjunctivitis. Other ocular findings include lid margin and conjunctival telangiectasias, eyelid thickening, eyelid crusts and scales, chalazia and hordeolum, punctate epithelial erosions, corneal infiltrates, corneal ulcers, corneal scars, and vascularization. Sight-threatening disease is rare with rosacea; however, keratitis can result in sterile corneal ulceration and eventual perforation if not treated aggressively.[1, 2, 3, 9, 10]

Ocular rosacea is most frequently diagnosed when patients also suffer from cutaneous disease. However, ocular signs and symptoms may occur prior to cutaneous manifestations in 20% of patients with rosacea. No correlation exists between the severity of ocular disease and the severity of facial rosacea.

The symptoms of rosacea can be treated effectively; however, rosacea is a chronic condition with exacerbations and remissions, which requires long-term therapy to maintain symptomatic control.

Pathophysiology

The precise pathophysiology of rosacea remains unclear[11, 12] but comprises both vascular dysregulation and altered immune system responses and inflammatory changes.[13] Recent research has shown an upregulation of proinflammatory and vasoregulatory genes in rosacea patients. Alterations in the innate immune system responses include an overabundance of cathelicidin (an antimicrobial peptide), along with kallikrein-5, an enzyme involved in processing cathelicidin. Moreover, toll-like receptor 2 activity in the innate immune system is increased in patients with rosacea.[8]

A variety of rosacea triggers have been described including skin colonization with Demodex mites (along with bacteria in their gut)[14] and Staphylococcus epidermidis.[15] Eradication of Helicobacter pylori has been shown to improve rosacea in some patients, and the organism may play a role in the pathogenesis of inflammation in rosacea.[2]

Four distinct rosacea subtypes have been described: erythematotelangiectatic rosacea, papulopustular rosacea, phymatous rosacea, and ocular rosacea.

Epidemiology

Frequency

United States

More than 10% of the general population exhibits dermatologic characteristics of rosacea; of these, up to 60% experience ocular complications.

International

An epidemiological study in Sweden showed a 10% prevalence of rosacea.[16] A study in Estonia showed a 22% prevalence rate of rosacea, as determined by the American National Rosacea Society Expert Committee (NRSEC) classification.[17]

Mortality/Morbidity

While rosacea is not a life-threatening disease, it is a source of much morbidity because of pruritus, burning, and psychosocial impairments. Approximately 5% of patients with rosacea manifest corneal disease, which can rarely be severe and lead to blindness via corneal ulceration, perforation, secondary infections, or corneal opacification from complete vascularization.

Race

Rosacea is recognized much more commonly in fair-skinned, white patients than in dark-skinned patients. However, because dark skin tones may mask erythema of rosacea, its incidence in this population is likely underreported.

Sex

Women are affected with the papulopustular and erythematotelangiectatic rosacea subtypes twice as often as men; however, the phymatous rosacea subtype develops much more frequently in men. Ocular rosacea affects both sexes equally.

Age

All ages can be affected, including pediatric patients.[18] Peak incidence occurs in the fourth to seventh decades of life.

Young children with chalazion may have ocular rosacea but may be uncooperative for slit-lamp examination.

Prognosis

Rosacea can be controlled symptomatically but generally is a chronic condition, which requires long-term, follow-up care.

Patient Education

Informing patients of the chronic, relapsing nature of rosacea is important so that patient expectation matches available therapy and patient follow-up care is maximized.

Ophthalmologists may underdiagnose rosacea because of a lack of familiarity with the dermatologic manifestations of the disease.

Presentation

History

Facial symptoms may include the following[1, 2, 3] :

- Recurrent flushing episodes
- Persistent and/or recurrent midfacial erythema
- Papular and pustular lesions

Ocular symptoms may include the following[1, 2, 3] :

- Dry eyes,[19] irritation, redness, itching, burning, foreign body sensation, and photophobia
- Recurrent chalazia
- Recurrent eye infections

Physical

Facial findings

Facial findings are as follows:

- Telangiectasias
- Papules and pustules (without comedones)
- Rhinophyma (hypertrophy of sebaceous glands leading to bulbous deformity of the nose)

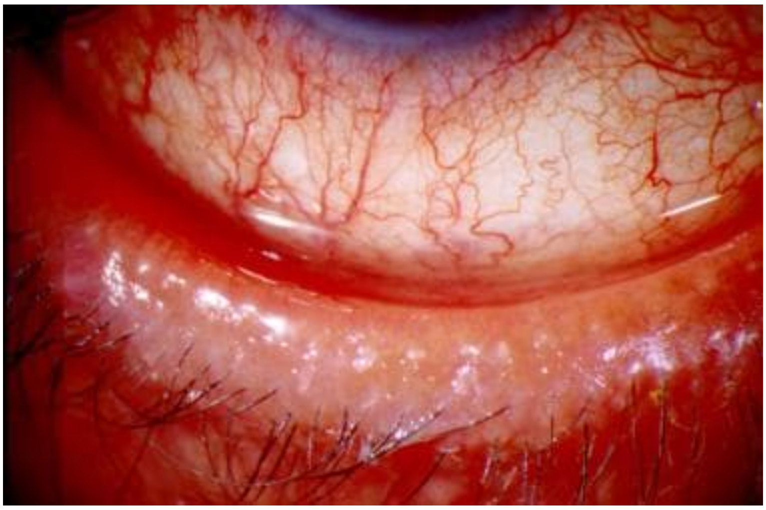
See the image below

Typical findings of rosacea, including papules, pustules, and rhinophyma.

Ocular findings

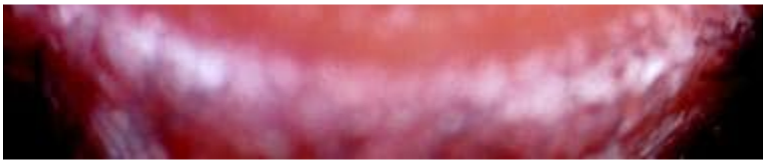
Ocular findings are as follows:

- Eyelid (most common)[20]
  - Eyelid telangiectasias as depicted below



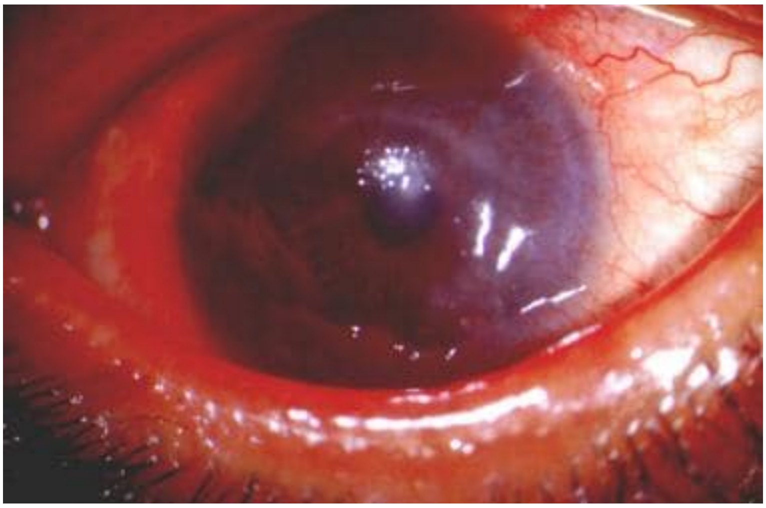
Ocular rosacea. Eyelid telangiectasias with inspissated meibomian glands.

  - Blepharitis
  - Meibomian gland dysfunction
  - Thick viscous plugging of meibomian gland orifices
  - Hordeola/chalazia
- Conjunctivitis
  - Usually chronic, diffuse hyperemia
  - Can lead to cicatrization in rare, severe cases
- Corneal findings
  - Punctate epithelial keratopathy (PEK), usually in the inferior one third of the cornea
  - Marginal corneal infiltrates
  - Corneal neovascularization
  - Superficial, wedge-shaped (spade-like) peripheral vascularization with its base at the limbus
  - Can progress to frank corneal neovascularization and eventual opacification as shown below



Ocular rosacea. Extensive corneal neovascularization and opacification.

- ○ Corneal thinning as depicted below, ulceration, and perforation



Ocular rosacea. Extensive corneal pannus with thinning.

- Secondary bacterial keratitis
- Episcleritis, scleritis (rare)

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Causes

Flushing triggers: These include alcohol, hot beverages, tobacco, spicy foods, vasodilating medications, and emotional stress.

UV light: This is postulated to decrease the competence of already dilated cutaneous vasculature, increasing persistent erythema and telangiectasias.

Demodex: This mite, which is part of the skin's normal flora, leads to stimulation of the innate immune system. Bacteria in the gut of Demodex may be the inciting factor rather than the Demodex itself.[14]

S epidermidis: Hyper-reactivity of the innate immune system in rosacea patients makes them sensitive to this normal skin flora.[15]

H pylori: This is postulated to be strongly correlated with rosacea. This is possibly due to a flush-inducing toxin present in H pylori.

Positive family history: Some studies have shown a higher rate of positive family history of rosacea in patients with this dermatologic disorder than in skin-healthy controls.[21]

Smoking: Some studies have shown an increased history of smoking in patients with rosacea as compared with skin-healthy controls.[21, 22]

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Complications

Complications include corneal vascularization, ulceration, perforations, secondary bacterial infections, and, ultimately, decreased vision.

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DDx

Differential Diagnoses

- Adult Blepharitis
- Allergic Conjunctivitis
- Atopic Dermatitis in Emergency Medicine
- Atopic Keratoconjunctivitis (AKC)
- Bacterial Conjunctivitis (Pink Eye)
- Bacterial Keratitis
- Central Sterile Corneal Ulceration
- Chalazion
- Chlamydia (Chlamydial Genitourinary Infections)
- Cicatricial (Mucous Membrane) Pemphigoid
- Corneal Ulcer
- Dry Eye Disease (Keratoconjunctivitis Sicca)
- Episcleritis
- Keratoconjunctivitis, Sicca
- Recurrent Corneal Erosion
- Viral Conjunctivitis (Pink Eye)

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Workup

Workup

Laboratory Studies

The diagnosis of ocular rosacea is established clinically and often is aided by dermatologic findings. Laboratory studies are not indicated. In cases of neovascularization and atypical presentation, ocular surface squamous neoplasia (OSSN) should be excluded.

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Imaging Studies

Imaging studies are not indicated.

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Other Tests

Other tests are not indicated.

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Histologic Findings

The conjunctiva in ocular rosacea is infiltrated by inflammatory cells, T-helper/T-suppressor (CD4) cells, phagocytic cells, and antigen-presenting cells. In addition, increased vascular dilation and occasionally granulomatous changes are present. None of these changes are specific for rosacea.

Patients with rosacea typically have a mean increase of nearly all cell types, but especially T-helper cells.

Hoang-Xuan et al demonstrated a 3.5-fold increase in the ratio of CD4 cells to CD8 cells in the conjunctiva of patients with rosacea, most resembling a type IV hypersensitivity reaction.[23]

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Treatment

Medical Care

Rosacea is caused by inherent defects in the body's immune system and vasoregulatory processes. Treatment is directed toward symptomatic control and disease prevention rather than cure.

When treating ocular rosacea, a stepwise approach can be undertaken, using first lid hygiene and artificial tears, followed by topical and oral anti-inflammatory medications, with late surgical intervention as required. Contact lenses should be avoided until the ocular rosacea is controlled.

Lid hygiene

Warm compresses applied to the eyelid margins can help to liquefy the thick meibomian gland secretions and, thus, facilitate their expression, although excessive heat application can exacerbate symptoms of eyelid irritation. Mild, nonirritating cleaning solutions, such as dilute baby shampoo or commercially prepared eyelid scrubs, also can be applied to the eyelids to remove clogging debris. Additionally, light pressure applied to the eyelids can aid in gland expression. Thermal pulsation to the eyelid (Lipiflow, Intense Pulse Light) is a technique used in the treatment of blepharitis. BlephEx is another emerging technique for the treatment of blepharitis using a motorized hand held instrument to debride accumulated debris from the eylid margins. Some studies have shown improvements in Demodex colony counts and inflammation with the use of 50% tea tree oil scrubs.[24, 25]

Artificial tears

Because of the frequency of application, nonpreserved artificial tears are recommended for use. Tears should be applied liberally throughout the day, and, if necessary, a lubricating ointment may be used at night. This ointment may contain an antibiotic preparation. Rosacea often results in evaporative dry eye due to a deficient oil layer on the surface of the tear film, so some patients benefit from lubricant eyedrops containing an oily emulsion. Miebo ™ (100% perfluorohexyloctane ophthalmic solution) is available by prescription as a perservative free eyedrop that reestablishes the oil layer and improves vision and ocular comfort.[26]

Immunosuppressive agents

Cyclosporine (cyclosporine ophthalmic) inhibits various T-cell activities, and topical formulations of cyclosporine have been successfully used for ophthalmic inflammatory conditions. Twice-daily instillations of topical cyclosporine have proven more efficacious than artificial tears in the management of the ocular surface changes and subjective symptoms of ocular rosacea.[27]

Antibiotics

Patients with ocular rosacea who are asymptomatic and without worsening eye disease should not be placed on oral antibiotics.

Tetracyclines (eg, tetracycline, doxycycline, minocycline)[28, 29, 30]

Tetracyclines represent the most common and most effective treatment regimen for rosacea. These drugs are believed to be effective not primarily as antibiotics but rather through a secondary effect that they exert on the meibomian glands. Tetracyclines decrease bacterial lipase, thereby altering the fatty acid composition of the meibomian gland secretions and improving their solubility. These medications also inhibit collagenase; therefore, they are effective in protecting the cornea from impending perforation secondary to inflammatory responses.

Adverse effects are predominantly GI, including diarrhea and, rarely, pancreatitis and pseudomembranous colitis. In these patients, enteric-coated tetracyclines such as Doryx (a form of enteric-coated doxycycline) are a promising option. The special coating prevents the medication from dissolving in the stomach where it may induce GI upset. Instead, the medication is broken down in the small intestine from where it readily enters the blood stream. More severe but much less common adverse effects include benign intracranial hypertension and renal tubular damage (Fanconi syndrome) from outdated medications. Additionally, tetracyclines cross the placenta and can cause permanent discoloration of teeth as well as fetal bone growth retardation.

Tetracyclines generally are effective for rosacea in doses much lower than those given for antibiotic effect, and, once the disease has come under control, the dose may be tapered to a lower, suppressive dose and maintained indefinitely. Because of the chronic, relapsing nature of rosacea, the medication may be used chronically at suppressive doses or discontinued and restarted if and when symptoms recur.

Among this class of medications,tetracycline and doxycycline most commonly are used. These two medications are quite similar in their mechanism of action, adverse effect profile, and efficacy, but slight differences do exist. Tetracycline has a shorter half-life and, thus, is dosed 4 times per day, as opposed to doxycycline, which is given twice per day or once per day. Frucht-Pery et al reported a more rapid therapeutic response to tetracycline; however, no difference was found at 6 months.[18]

In 2006, the first FDA-approved oral treatment for rosacea became available: a controlled-release form of doxycycline called Oracea (Galderma Laboratories LP). The 40-mg tablet is a combination of 30 mg of immediate-release and 10 mg of delayed-release doxycycline. The low dose enables the medication to have anti-inflammatory properties without exerting significant antibacterial properties, allowing for a more improved side effect profile and decreased rates of bacterial resistance.

Topical azithromycin eye drops also have gained popularity in the treatment of ocular rosacea. Ocular rosacea often results in severe and recalcitrant blepharitis. Azasite (azithromycin ophthalmic 1%, Inspire Pharmaceuticals), currently FDA approved only to treat bacterial conjunctivitis, has found an off label use in the treatment of meibomian gland dysfunction.

Erythromycin can be taken orally for patients intolerant to, or too young for, tetracyclines. Erythromycin ointment applied to the lid margins once or twice daily can provide lubrication for the eye and reduce the bacterial overgrowth contributing to lid margin disease.

Clarithromycin has shown efficacy in treating rosacea. This compound exhibits anti-inflammatory effects as well as activity against H pylori. Torresani compared clarithromycin and doxycycline and found equivalent therapeutic responses and a milder adverse effect profile for clarithromycin.[31]

Metronidazole

Metronidazole exhibits antimicrobial (antibacterial and antiparasitic), anti-inflammatory, and immunosuppressive properties and has been found to be effective against rosacea. In fact, oral metronidazole has been advocated as first-line therapy. Adverse effects include gastrointestinal irritation and a disulfiramlike action; thus, abstinence from alcohol is required.

Topical metronidazole is quite effective in treating skin lesions in rosacea. While not approved for ophthalmic use, in a pilot study, Barnhorst et al found the topical compound to be safe and effective in treating eyelid involvement in ocular rosacea.[32]

Topical steroids

Topical steroids can prove useful for short-term exacerbations of lid disease and management of inflammatory keratitis. However, steroids should be used cautiously and discontinued as soon as possible to prevent corneal melting. Topical steroids may lead to rosacea exacerbations and should be avoided if possible.

Retinoids

Vitamin A derivatives, such as oral isotretinoin and topical tretinoin, have been found effective in reducing the inflammatory lesions in rosacea. This appears to be accomplished via the suppression of sebum production and a subsequent reduction in sebaceous follicle size. Additionally, tretinoin may help restore sun-damaged skin through the increased production of type 1 collagen in damaged regions. Both compounds actually can cause severe erythema and blepharoconjunctivitis, worsen telangiectasias, and lead to severe keratitis. Additionally, retinoids are extremely teratogenic and, thus, must never be used during pregnancy. Therefore, the use of retinoids commonly is reserved for cases in which multiple agents have failed.

Antiulcer therapy

H pylori plays an as yet undetermined role in rosacea, and some have advocated H pylori eradication in the treatment of rosacea. Thus, in some cases of rosacea, antiulcer combination regimens, such as amoxicillin or clarithromycin, metronidazole, bismuth, and an H2 antagonist, have been used with varying efficacy.

Topical acne treatments

Azelaic acid

Azelaic acid 15% gel (Finacea) has multiple modes of action in rosacea. It is an antibacterial agent that inhibits the growth of susceptible organisms (principally Propionibacterium acnes) on the surface of the skin by inhibiting protein synthesis, but an anti-inflammatory effect achieved by reducing reactive oxygen species appears to be the main pharmacological action.[33]

Ivermectin

Ivermectin 1% cream (Soolantra) has been approved by the US Food and Drug Administration since 2014 for the treatment of the inflammatory lesions characteristic of rosacea. Its mechanism of action is not clear but it decreases the density of Demodex mites found in the skin of people with rosacea when applied daily. [34]

Other treatments

Intradermal injections of botulinum toxin have been shown to reduce rosacea-associated facial erythemas for up to several months.[35]

Other treatments in the treatment of rosacea include intense pulsed light therapy, ablative lasers, and electrosurgical loop.[36]

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Surgical Care

Surgical care includes the following:

- Treatment of dry eye - Punctal occlusion can be accomplished via permanent silicone plugs or punctal cauterization.
- Amniotic membrane - Amniotic membrane has anti-inflammatory properties and promotes reepithelialization of the cornea. It can be used to reconstruct the corneal surface in severe cases of rosacea when a nonhealing epithelial defect, corneal ulceration, or limbal stem cell deficiency are present.
- Treatment of corneal perforations
    - Cyanoacrylate tissue adhesive
    - Lamellar keratoplasty
    - Penetrating keratoplasty
- Restoration of vision from corneal disease
    - Penetrating keratoplasty
    - The success rate for graft survival generally is much lower than for noninflammatory conditions because of the increased vascularization of the host cornea.
- Treatment of limbal stem cell deficiency - Limbal stem cell transplant

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Consultations

A dermatology consult is recommended for the optimal management of rosacea.

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Diet

Avoidance of triggers, such as hot, spicy foods, alcohol, and heated beverages, can reduce symptomatic episodes.

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Activity

Avoidance of sunlight can be beneficial for some patients.

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Complications

Eyes undergoing penetrating keratoplasty are more likely to experience graft rejection than eyes without rosacea because of the increased inflammatory response and the relatively increased corneal vasculature.

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Prevention

Patients should avoid trigger foods and situations.

For some patients, avoidance of sunlight can minimize flare-ups.

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Further Outpatient Care

Rosacea is a chronic condition, and long-term management is necessary to control this disease.

Dermatology and ophthalmology visits may be necessary, and they initially could be frequent to gain control over the symptoms or to protect an endangered cornea.

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Further Inpatient Care

Inpatient care rarely is necessary, except in some cases of corneal perforations or severe secondary corneal infections.

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Medication

Medication Summary

The goals of pharmacotherapy are to reduce morbidity and to prevent complications.

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Antibiotics

Class Summary

Anti-inflammatory effect helps to ameliorate meibomian gland disease.

Tetracycline

Inhibits bacterial protein synthesis by binding with 30S and possibly 50S ribosomal subunit(s). Has anti-inflammatory activity. Also a potent collagenase inhibitor.

Doxycycline (Vibramycin, Doryx, Adoxa, Alodox)

DOC; inhibits bacterial protein synthesis by binding with 30S and possibly 50S ribosomal subunit(s). Has anti-inflammatory activity. Also a potent collagenase inhibitor.

Clarithromycin (Biaxin, Biaxin XL)

Inhibits bacterial growth, possibly by blocking dissociation of peptidyl t-RNA from ribosomes causing RNA-dependent protein synthesis to arrest. Effective through secondary, anti-inflammatory action.

Metronidazole (Flagyl, MetroCream, Rosadan, Vandazole)

Has anti-inflammatory and immunosuppressive activity. Topical and systemic dosage forms have been found to be effective.

Erythromycin ophthalmic (Ilotycin)

Used to decrease meibomian gland bacterial overgrowth.

Azithromycin ophthalmic (AzaSite)

This ophthalmic macrolide antibiotic is indicated for bacterial conjunctivitis may be used off label in the treatment of meibomian gland dysfunction.

Retinoid-like Agents

Class Summary

Decrease sebaceous gland size and sebum production. May inhibit sebaceous gland differentiation and abnormal keratinization.

Isotretinoin (Amnesteem, Claravis, Myorisan, Sotrel)

Reduces sebum production and sebaceous follicle size.

Tretinoin topical (Avita, Retin-A, Retin-A Micro, Ronova)

Structurally related to vitamin A. Reduces sebum production and sebaceous follicle size. Makes keratinocytes in sebaceous follicles less adherent and easier to remove. May help restore sun-damaged skin. Long-term, low-dose therapy may be suitable for selected patients.

Inhibits microcomedo formation and eliminates lesions present. Available as 0.025%, 0.05%, and 0.1% creams. Available also as 0.01% and 0.025% gels.

Corticosteroids

Class Summary

Topical steroids occasionally are needed to help suppress inflammatory changes in the cornea.

Prednisolone acetate (Omnipred, Pred Mild, Pred Forte)

Decreases inflammation and corneal neovascularization. Suppresses migration of polymorphonuclear leukocytes and reverses increased capillary permeability.

Immunosuppressants

Class Summary

These agents regulate key regulatory steps responsible for inflammation.

Cyclosporine ophthalmic (Restasis)

Used to relieve dry eyes caused by suppressed tear production secondary to ocular inflammation. Thought to act as partial immunomodulator. Exact mechanism of action is not known.

Ophthalmic Lubricants

Class Summary

Lubricants act as humectants in the eye. The ideal artificial lubricant should be preservative-free; contain potassium, bicarbonate, and other electrolytes; and have a polymeric system to increase its retention time. Lubricating drops are used to reduce morbidity and to prevent complications. Lubricating ointments prevent complications from dry eyes. Ocular inserts reduce symptoms resulting from moderate to severe dry eye syndromes.

Artificial tears (Advanced Eye Relief, Bion Tears, Hypo Tears, Murine Tears, Tears Naturale II)

Artificial tears are used to increase lubrication of the eye. Nonpreserved artificial tears are recommended for use. Tears should be applied liberally throughout the day, and, if necessary, a lubricating ointment may be used at night. This ointment may contain an antibiotic preparation.

Acne Agents, Topical

Azelaic acid (Azelex, Finacea)

Azelaic acid 15% gel (Finacea) has multiple modes of action in rosacea. It is an antibacterial agent that inhibits the growth of susceptible organisms (principally Propionibacterium acnes) on the surface of the skin by inhibiting protein synthesis, but an anti-inflammatory effect achieved by reducing reactive oxygen species appears to be the main pharmacological action.

Ivermectin topical (Sklice, Soolantra)

Ivermectin 1% cream (Soolantra) has been approved by the US Food and Drug Administration since 2014 for the treatment of the inflammatory lesions characteristic of rosacea. Its mechanism of action is not clear but it decreases the density of Demodex mites found in the skin of people with rosacea when applied daily.

Antiglaucoma, Alpha Agonists

Brimonidine (Lumify)

Brimonidine tartrate ophthalmic solution 0.025% (available OTC as Lumify) can be applied twice daily to reduce redness caused by vasodilation.

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Ophthalmics, Other

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Perfluorohexyloctane ophthalmic solution applied up to QID can be effective in treating visual symptoms caused by evaporative dry eye resulting from impaired meibomian gland activity.

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Ophthalmics, VEGF Inhibitors

Ranibizumab intravitreal injection (Byooviz, Cimerli, Lucentis)

VEGF Inhibitors have been used "off label" as subconjunctival injections applied to reduce corneal neovascularization associated with rosacea.

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Questions & Answers

Overview

What is ocular rosacea?

What is the pathophysiology of ocular rosacea?

What is the prevalence of ocular rosacea in the US?

What is the global prevalence of ocular rosacea?

What is the mortality and morbidity associated with ocular rosacea?

What are the racial predilections of ocular rosacea?

What are the sexual predilections of ocular rosacea?

Which age groups have the highest prevalence of ocular rosacea?

Presentation

What are the facial signs and symptoms of ocular rosacea?

What are the ocular signs and symptoms of rosacea?

Which facial findings are characteristic of ocular rosacea?

Which ocular findings are characteristic of rosacea?

What causes ocular rosacea?

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What are the differential diagnoses for Ocular Rosacea?

Workup

What is the role of lab tests in the workup of ocular rosacea?

Treatment

Medications

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